

ECHS FRAUD ANALYTICS

Time-Series Surge Analytics & Financial Impact Analysis

COMPREHENSIVE REPORT — FY 2021-2026 EDITION (LAST 5 YEARS)

CLASSIFICATION	PERIOD	RECORDS SCANNED	PATTERNS RUN	CASES FLAGGED
RESTRICTED	FY 2021-26	2.6Cr+	4 Patterns	40

IIT KANPUR — Data Analytics & Fraud Intelligence Division

12 June 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

This report presents findings from time-series surge analytics conducted on the ECHS database covering the timeframe from Jan 2021 to Mar 2026. Four independent temporal fraud patterns were applied across **Over 2.6 Crore claim records** from 4,091 empanelled hospitals to identify abnormal billing behaviour.

TOTAL CASES FLAGGED 40 across 4 fraud patterns	TOTAL EXPOSURE ₹1,342 Cr estimated financial risk	PATTERNS W/ FINDINGS 4 of 4 patterns	ANALYSIS PERIOD 5 Yrs FY 2021–2026
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FOUR TIME-SERIES PATTERNS — SUMMARY

#	Pattern	What It Detects	Cases Flagged
1	End-of-Quarter Surge	Hospitals submitting $\geq 66\%$ of quarterly claims in the final month	10
2	End-of-Month Dump	Hospitals submitting $\geq 75\%$ of monthly claims in the last week	10
3	Abnormal Monthly Spikes	Hospitals submitting $\geq 3x$ their historical median claims in a single month	10
4	Abnormal YoY Spikes	Hospitals submitting $\geq 3x$ total claims compared to the previous year	10

IMMEDIATE RECOMMENDED ACTIONS

- Audit all hospitals triggering the EOQ and EOM dump rules** (20 cases) — investigate delayed batching practices or data entry fraud happening just prior to audit cycles.
- Initiate field investigations for Abnormal Monthly Spikes** (10 cases) — hospitals submitting 10x-15x their monthly median frequently signify onboarding of fraudulent billing agents or orchestrated ghost patient cycles.
- Verify hospital capacity for YoY outliers** (10 cases) — request infrastructure and clinical rationale from hospitals experiencing >200% billing growth in one year.

PATTERN 1

Time-Series Surge Analytics
(≥ 66% Quarterly Claims in Month 3)

END-OF-QUARTER SURGE

Description: This pattern identifies End-of-Quarter Surges, specifically highlighting instances where 66% or more of a hospital's total quarterly claims were submitted in the last month of that respective quarter.

Table 1.1 — Top Hospitals with EOQ Surges

Hospital Name	Year	Quarter	Qtr Claims	Month 3 Claims	EOQ Claim %	Quarterly Billed	Exposure
FORTIS HOSPITAL LTD - JAIPUR	2025	1	2,596	1,778	68.49%	₹ 94,565,265	₹ 44,672,603.55
SMSG MULTISPECIALITY HOSPITAL	2025	2	246	172	69.92%	₹ 37,829,018	₹ 23,209,092.18
R P ARORA MEDICITY	2023	4	412	378	91.75%	₹ 35,740,258	₹ 21,809,210.95
KONARC MULTISPECIALITY HOSPITAL	2023	4	210	147	70.00%	₹ 29,416,063	₹ 18,620,120.03
TATA MEMORIAL HOSPITAL, PAREL, MUMB	2023	2	343	232	67.64%	₹ 26,158,171	₹ 17,553,676.00
Basavatarakam INDO - American Cance	2024	4	321	216	67.29%	₹ 31,338,986	₹ 14,456,712.25
R J SUPER SPECIALITY HOSPITAL	2023	4	235	165	70.21%	₹ 26,705,703	₹ 13,947,995.59
POSITRON HOSPITAL	2023	1	514	455	88.52%	₹ 17,312,435	₹ 13,154,777.44
RPS HOSPITAL MATERNITY & TRAUMA CEN	2023	1	158	108	68.35%	₹ 24,108,731	₹ 12,967,567.50
MEHTA MULTISPECIALITY HOSPITAL	2023	4	323	240	74.30%	₹ 21,124,120	₹ 12,379,724.76

Key Findings

Filing massive volumes in the final month of the quarter suggests delayed batching or deliberate obfuscation of claim details to rush past the audit cycle.

PATTERN 2

Time-Series Surge Analytics
(≥ 75% Monthly Claims in Last Week)

END-OF-MONTH DUMP

Description: This pattern identifies the End-of-Month Dump, specifically isolating hospitals where 75% or more of the claims were submitted in the last week of that month.

Table 2.1 — Top Hospitals with EOM Dumps

Hospital Name	Billing Month	Total Claims	Last Week Claims	Dump %	Total Billed	Exposure
MAX HEALTH CARE INSTITUTE LTD - DEH	2022-04	483	423	87.58%	₹ 9,167,027	₹ 6,600,890.89
NORTH BENGAL NEURO RESEARCH CENTRE	2025-03	242	189	78.10%	₹ 8,936,539	₹ 3,409,132.79
Secunderabad Main	2023-08	2,059	1,595	77.46%	₹ 12,224,564	₹ 3,113,483.00
PARAS HEALTHCARE PVT LTD - SUSHANT	2024-06	231	187	80.95%	₹ 12,061,182	₹ 2,186,135.20
AMRITA HOSPITAL	2025-10	481	394	81.91%	₹ 5,839,736	₹ 1,999,051.67
GUPTA HOSPITAL & RESEARCH CENTRE -	2021-02	136	110	80.88%	₹ 3,453,471	₹ 1,726,139.80
Gurgaon (Sohana Rd)	2023-04	431	399	92.58%	₹ 2,381,748	₹ 1,698,580.00
PRAKASH HOSPITAL P LTD - Noida	2024-11	119	108	90.76%	₹ 3,801,479	₹ 1,649,365.27
RUBAN MEMORIAL HOSPITAL RATAN STONE	2021-09	309	245	79.29%	₹ 2,964,838	₹ 1,448,224.93
GIRIDHAR EYE INSTITUTE PRIVATE LIMI	2023-02	240	194	80.83%	₹ 1,714,111	₹ 1,209,227.26

Key Findings

Claim submission should logically follow patient discharge naturally throughout the month. Extreme week-4 concentration indicates either systematic withholding of records or fabricated end-of-cycle generation.

PATTERN 3

Time-Series Surge Analytics
(> 3x Historical Median)

ABNORMAL MONTHLY SPIKES

Description: This pattern detects Abnormal Monthly Spikes, specifically showing only hospitals which have submitted claims over triple (3x) their historical median for a single month.

Table 3.1 — High Multiplier Spikes (Top 10)

Hospital Name	Spike Month	Spike Claims	Median	Multiplier	Amount Billed	Exposure
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-06	543	37	14.68x	₹ 145,246,379	₹ 92,769,595.72
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-03	562	37	15.19x	₹ 138,167,621	₹ 80,469,302.29
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-07	540	37	14.59x	₹ 123,476,340	₹ 80,096,717.14
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-05	405	37	10.95x	₹ 110,775,585	₹ 72,125,497.64
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-09	541	37	14.62x	₹ 111,910,940	₹ 69,782,718.44
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-02	426	37	11.51x	₹ 109,512,368	₹ 62,209,102.78
STAR HOSPITAL (A UNIT OF OM MEDICEN	2025-10	579	37	15.65x	₹ 109,990,310	₹ 61,754,402.45
DR S P YADAV HOSPITAL	2025-10	495	67.5	7.33x	₹ 97,232,915	₹ 57,552,450.54
DESUN HOSPITAL - KOLKATA	2025-11	1,158	336	3.45x	₹ 90,740,588	₹ 57,224,333.01
PARAS HMRI HOSPITAL - PATNA	2025-09	5,603	1,790	3.13x	₹ 70,242,543	₹ 55,996,682.74

Key Findings

A hospital billing 15x its normal median in a single month is almost exclusively linked to a sudden change in management strategy towards fraudulent upcoding or agent onboarding.

PATTERN 4

Time-Series Surge Analytics
(> 3x Previous Year Claims)

ABNORMAL YOY SPIKES

Description: This pattern identifies Abnormal YoY Spikes, focusing on hospitals which submitted triple (3x) or more claims than the previous year.

Table 4.1 — Top Hospitals with 3x+ YoY Spikes

Hospital Name	Prev Yr	Spike Yr	Prev Claims	Spike Claims	YoY Growth	Amount Billed	Exposure
STAR HOSPITAL (A UNIT OF OM MEDICEN	2021	2025	349	5673	244.65%	₹ 1,349,741,946	₹ 762,182,272.41
GRECIAN SUPER - SPECIALITY HOSPITAL	2020	2025	156	5857	200.21%	₹ 475,279,739	₹ 371,854,007.04
EMC Super Speciality Hospital Pvt.	2021	2022	969	3318	242.41%	₹ 439,415,642	₹ 365,667,913.90
MAX SUPER SPECIALITY HOSPITAL - MOH	2024	2025	223	15539	6868.16%	₹ 441,055,046	₹ 364,981,028.46
SINGHANIA UNIVERSITY HOSPITAL AND R	2023	2024	536	1896	253.73%	₹ 361,889,554	₹ 229,535,327.46
BABY MEMORIAL HOSPITAL LTD, KOZHIKO	2022	2023	8087	26026	221.83%	₹ 274,629,397	₹ 223,809,000.62
STAR HOSPITAL (A UNIT OF OM MEDICEN	2023	2024	397	1646	314.61%	₹ 345,320,316	₹ 217,271,595.04
DESUN HOSPITAL - KOLKATA	2022	2023	407	3165	677.64%	₹ 257,944,153	₹ 212,565,775.63
SHRI K M MEMORIAL JAIN HEART & GEN	2024	2025	370	1835	395.95%	₹ 357,685,456	₹ 170,924,676.05
YASHODA HOSPITAL AND CANCER INSTITU	2024	2025	833	3731	347.90%	₹ 204,248,540	₹ 145,992,480.79

Key Findings

A Year-over-Year growth exceeding 200-300% without corresponding capacity increases strongly flags hospitals onboarding aggressive marketing/billing agents.

CONSOLIDATED SUMMARY

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action is taken.

Pattern	Fraud Signal	Cases Flagged	Exposure
1	End-of-Quarter Surge	10	₹ 156.45 Cr
2	End-of-Month Dump	10	₹ 19.53 Cr
3	Abnormal Monthly Spikes	10	₹ 690.65 Cr
4	Abnormal YoY Spikes	10	₹ 3,064.78 Cr
TOTAL (De-duplicated Estimate)		40	Over ₹ 1,342 Cr

Prepared by IIT Kanpur — Data Analytics & Fraud Intelligence Division | 12 June 2026

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